

7. (a) Explain what is meant by the term 'multiwavelength astronomy'. [2]

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- (b) The Stefan constant, σ , has the unit $\text{W m}^{-2} \text{K}^{-4}$. Express this in terms of base SI units. [2]

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- (c) The wavelength of the peak emission of the bright star Arcturus is measured to be 674 nm.

- (i) Calculate the star's temperature. [2]

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- (ii) The visible spectrum extends from approximately 400 nm to 700 nm. State the colour of Arcturus. [1]

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- (d) A website claims that Arcturus is about 37 light years from the Sun. Use the following information and your answer to (c)(i) to justify this claim. [5]

Diameter of Arcturus = $2.78 \times 10^{10} \text{ m}$

Intensity of radiation received on Earth from Arcturus = $3.09 \times 10^{-8} \text{ W m}^{-2}$

1 light year = $9.46 \times 10^{15} \text{ m}$

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